

Preference Elicitation for Movie Recommendation

Human-Centric Artificial Intelligence — Project 4

0. Dataset note

The original submission uses the IMDB 5000 Movie Dataset. For the purposes of building and testing the full pipeline described below, a small hand-curated stand-in dataset (59 well-known films) was constructed with the identical column schema (movie_title, genres, director_name, actor_1/2/3_name, duration, title_year, imdb_score, language, country, content_rating, budget, gross, num_voted_users). Every function in the feature-extraction and modeling code operates purely on these column names, so pointing the pipeline at the real movie_metadata.csv file requires no code changes — only re-running the extraction script.

1. Feature representation (Task 1)

The preference model assumes a linear utility $U(x) = w^T x$. This places two requirements on the feature map x : (a) it must be low-dimensional and dense enough that w can be estimated from a handful of interactions (the whole point of the project is fast adaptation), and (b) it must be computable for every movie in the catalogue, including ones never shown during elicitation, so a fitted w can score and rank the full catalogue afterwards.

Three raw fields are especially problematic for a linear model: director and the three lead actor names. A one-hot encoding over these would add hundreds of near-empty dimensions (many directors/actors appear once), guaranteeing overfitting with the amount of data one user can realistically provide. Instead we use leave-one-out target encoding: each director/actor is replaced by the average IMDB score of that person's other movies in the corpus (falling back to the corpus mean when the person appears only once). This keeps the whole cast/crew signal in two scalar dimensions.

Final feature vector (27 dimensions on the stand-in dataset)

Group	Feature(s)	Encoding	Rationale
Genre	genre::Drama, genre::Comedy, ... (top-K genres)	Multi-hot (0/1), one dim per genre	Genre is the single strongest, most stable driver of taste and generalizes across the whole catalogue
Runtime	duration	Min-max normalized to [0,1]	Captures preference for shorter vs. longer films
Era	recency	Normalized release year	Captures preference for classic vs. contemporary films
Popularity	popularity	$\log(1+\text{num_voted_users})$, normalized	Distinguishes mainstream blockbusters from niche titles
Quality prior	quality_prior	Normalized IMDB score	A crowd-level quality signal, useful as a prior even though it is not the personal utility we estimate
Crew/cast	director_prestige, cast_prestige	Leave-one-out mean IMDB score of the person's other films	Compact (2-dim) stand-in for hundreds of sparse director/actor one-hot dimensions
Content	maturity	Ordinal (G<PG<PG-	Captures tolerance for

		13<R<NC-17), normalized	mature content
Origin	is_english, is_domestic	Binary	Captures foreign-film / arthouse taste

Extraction code (features.py) computes this vector for every row of the dataset in a single vectorized pass and was validated on the 59-movie stand-in set (output: 27-dimensional feature matrix, one row per film).

2. Extending Bradley-Terry to rankings (Task 2)

The standard Bradley-Terry model gives the probability that item i is preferred to item j as:

$$P(i \succ j \mid w) = \exp(w \cdot x_i) / [\exp(w \cdot x_i) + \exp(w \cdot x_j)]$$

Design 2 of the study asks participants to rank ten movies rather than compare two, so we need $P(i_1 \succ i_2 \succ \dots \succ i_n \mid w)$. We adopt the Plackett-Luce model, which generates a full ranking as a sequence of Bradley-Terry-style draws from a shrinking pool: first the most-preferred item is chosen from all n candidates with probability proportional to $\exp(w \cdot x)$; it is removed from the pool; the next item is chosen from the remaining $n-1$ with the same rule; and so on until one item remains.

$$P(i_1 \succ i_2 \succ \dots \succ i_n \mid w) = \prod_{k=1}^n \exp(w \cdot x_{i_k}) / \sum_{j=k}^n \exp(w \cdot x_{ij})$$

This is a strict generalization: setting $n = 2$ collapses the product to a single term that is exactly the pairwise Bradley-Terry probability above. Consequently both study designs are special cases of the same underlying model class, which is what makes it meaningful to compare them: every trial, from either interface, is simply "one more ranking observation" (a pairwise choice being a ranking of length 2), and the same log-likelihood, in the same units, can be computed for data collected under either design.

We fit w by maximum likelihood, i.e. minimizing the negative log-likelihood above (summed over all elicited rankings) via gradient descent, with a small L2 penalty acting as a Gaussian prior on w . The penalty matters specifically because the whole premise of the project is estimating w from very few interactions, where an unregularized MLE would be poorly conditioned and prone to wild extrapolation onto unseen movies. This was implemented (preference_model.py) and validated on synthetic data: a simulated user with a strong preference for Animation and an aversion to Horror was recovered correctly from 40 simulated pairwise comparisons, with the two largest-magnitude fitted weights landing on exactly those two genres.

3. User study design (Task 3)

3.1 Research questions and hypotheses

The study compares two preference-elicitation interfaces under a single underlying preference model (Section 2), asking which one elicits a user's preferences more effectively per unit of participant effort.

- H1 (accuracy): Full-ranking elicitation (Design 2) yields a preference vector w with higher predictive accuracy on held-out pairwise comparisons than pairwise elicitation (Design 1), for the same number of elicitation trials, because a ranking of 10 items implies $C(10,2)=45$ pairwise constraints per trial versus 1 for Design 1.
- H2 (effort/time): Design 2 trials take longer and are perceived as more cognitively demanding per trial than Design 1 trials, because ranking ten items requires holding more information in mind than a single binary choice.
- H3 (efficiency, accuracy per second): It is not a priori clear which design wins once time is accounted for; we test whether Design 1's speed offsets Design 2's richer signal, i.e. whether accuracy-per-minute differs between conditions.
- H4 (subjective preference): Participants report higher satisfaction / lower perceived workload for one interface over the other (exploratory, no directional prediction).

3.2 Design

Between-subjects design (each participant experiences only one elicitation method) to avoid carry-over/fatigue effects and ordering confounds that a within-subjects design would introduce given that both conditions elicit the same

underlying construct. Random assignment to condition (Design 1 vs. Design 2), stratified by self-reported movie-watching frequency (light/moderate/heavy) collected at intake, to balance a plausible confound across conditions.

Factor	Design 1 (pairwise)	Design 2 (ranking of 10)
Independent variable	Elicitation method	Elicitation method
Trials	20 pairwise choices	4 rounds of ranking 10 movies
Movie exposures	40 (2×20)	40 (10×4)
Implied pairwise constraints	20	180 (45×4)
Movie sampling	Uniform random from catalogue, no repeats within a session	Uniform random from catalogue, no repeats within a session

Movie exposure count is matched across conditions (40 films shown either way) so that any accuracy difference is attributable to the elicitation format rather than to participants simply seeing more movies.

3.3 Procedure

- 1. Landing page and informed consent (purpose, data usage, voluntary participation, right to withdraw, anonymized data).
- 2. Brief intake questionnaire: age bracket, movie-watching frequency, familiarity with recommender systems.
- 3. Random assignment to Design 1 or Design 2 (stratified on watching frequency).
- 4. Elicitation task in the assigned condition, with an unobtrusive on-screen timer.
- 5. Common held-out evaluation block, identical for all participants regardless of condition: 10 pairwise comparisons between movies not shown during elicitation. Used purely to measure predictive accuracy of the just-fitted w ; not used in fitting.
- 6. Post-task questionnaire: perceived ease of use and mental effort (5-point Likert, adapted from NASA-TLX mental-demand item), enjoyment, and a forced-choice "which style would you prefer for a real recommender app."
- 7. Debrief screen showing the participant their own top-5 personalized recommendations from the fitted model, and a thank-you/compensation note.

3.4 Measures (dependent variables)

Measure	How computed	Relates to
Predictive accuracy	% of the 10 held-out pairwise comparisons correctly predicted by $\text{sign}(w \cdot x_i - w \cdot x_j)$	H1
Elicitation time	Wall-clock seconds from first trial shown to last trial submitted	H2
Accuracy per minute	Predictive accuracy / elicitation time	H3
Perceived workload	5-point Likert, mental-demand item	H2, H4
Satisfaction / preference	5-point Likert + forced-choice item	H4

3.5 Sample size / power analysis

Using a two-sample t-test for the primary comparison (predictive accuracy between conditions), targeting $\alpha = .05$ (two-tailed) and power = .80 to detect a medium effect size (Cohen's $d = 0.5$, a conservative but realistic expectation for an interface-design manipulation in HCI), the standard formula $n = 2(z_{1-\alpha/2} + z_{1-\beta})^2 \sigma^2 / \delta^2$ gives approximately $n \approx 63$ participants per condition. We therefore target 64 per condition (128 total), rounding up and adding an expected 10% attrition/exclusion buffer, i.e. recruit approximately 140 participants.

3.6 Recruitment

- Population: adults (18+) with normal or corrected-to-normal vision able to complete a ~10-minute online task; no requirement of being a frequent moviegoer, since the recommender is meant to work for casual and heavy users alike (intake question used only for stratification, not exclusion).
- Channel: a crowdsourcing platform (e.g. Prolific) for a demographically broad, paid sample, supplemented by university mailing lists for a pilot batch.

- Compensation: a fixed fee corresponding to a fair hourly rate for the estimated ~10 minute completion time, paid regardless of the participant's answers (to avoid biasing responses toward what participants think the researcher wants).
- Exclusions (pre-registered): completion time under 90 seconds (likely inattentive), and an attention-check trial (a pair with one obviously non-existent/placeholder "movie") failed.

3.7 Ethics

- Informed consent obtained before any data collection; participants can withdraw at any point without penalty or loss of already-earned compensation for completed sections.
- Data collected is limited to task responses, timing, and the coarse intake questionnaire; no directly identifying information is collected beyond what the recruitment platform itself requires for payment.
- Movie stimuli are limited to publicly available metadata (titles, genres, cast/crew, run time); no video/audio clips or copyrighted stills are shown.
- Institutional review/ethics board approval would be sought before running the study on real participants, per standard human-subjects research practice.

3.8 Analysis plan

- Primary: independent-samples t-test (or Mann-Whitney U if normality is violated) on predictive accuracy between conditions (H1).
- Secondary: t-tests on elicitation time and accuracy-per-minute (H2, H3); ordinal regression or Mann-Whitney U on Likert workload/satisfaction items (H4).
- Exploratory: regression of predictive accuracy on condition, watching-frequency stratum, and their interaction, to check whether one method is disproportionately better for a particular user segment.
- All tests two-tailed at $\alpha = .05$; effect sizes (Cohen's d or rank-biserial correlation) reported alongside p-values.

4. Implementation (Task 4)

The accompanying interactive project page implements the full pipeline described above:

- Landing page: study description, consent text, a button to download this report (PDF), and a "Start Study" button.
- Random, stratified condition assignment on entering the study.
- Design 1 interface: sequential pairwise choice screens (20 trials).
- Design 2 interface: drag-to-reorder ranking screens, ten movies at a time (4 rounds).
- A shared held-out evaluation block (10 pairwise trials) shown to all participants after the assigned condition.
- Client-side model fitting: the Plackett-Luce maximum-likelihood estimator from Section 2, re-implemented in JavaScript, runs in-browser on the collected trials to fit w .
- Post-task questionnaire and a personalized debrief screen (top-5 recommended movies from the fitted w).
- All trial-level data, timings, and questionnaire responses are persisted so that, were the study actually run, every participant's full record (condition, choices, timings, fitted w , held-out accuracy, questionnaire answers) would be available for the analysis plan in Section 3.8.

The feature extraction (`features.py`) and model fitting (`preference_model.py`) scripts, together with the stand-in dataset generator (`make_sample_dataset.py`), are included alongside this report and are written so that pointing them at the real IMDB 5000 movie_metadata.csv reproduces the same pipeline end-to-end on the full 5000-movie catalogue.